ISAForge / how our area model works

A tree ensemble turns an instruction formula into an estimate of isolated FPGA logic cost.

RETAINED MODEL · V9 · 1,713 labeled originals

THE PREDICTION PATH

1 Formula

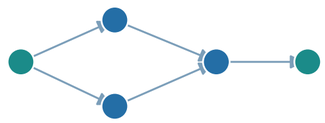
RV32 computation
with shared inputs

```
add(
  mul(in0,addi(0,341)),
  in1)
```

Only the formula enters
feature extraction.
No observed area or residual.

2 Graph views

Yosys generic lowering
before target ABC mapping



Original DAG
Normalized DAG
SOG structural descriptors

3 Fixed features

217 selected columns
from 336 computed

14 SOG
60 Original graph
60 Normalized graph
60 Absolute graph differences
23 Mapping descriptors

4 Tree ensemble

ExtraTreesRegressor



300 randomized trees
Raw LUT4 training target
Average tree predictions

5 Area estimate

Predicted LUT4

Isolated combinational
iCE40 datapath area

`max(0, forest mean)`

Original generic gates =0
forces the returned point to 0

Point estimate · uncalibrated

fit

TRAINING LABELS

Verified original RTL → **Yosys iCE40 synthesis** → **Count original LUT4**

V13 batch: 48 originals × 2,148 sample comparisons; all 48 original labels independently audited.

HOW WE IMPROVE IT / MEASURE, COMPARE, THEN DECIDE

1 Diagnose a mechanism

Correlated high products
Affine chains
Comparator/shift compositions

2 Add 48 training originals

Fixed before measurements
Verified original-RTL targets
All outcomes and failures retained

3 Run a matched comparison

538 jointly closed groups
5 folds × 3 seeds × 2 variants
30 fits; shared held-out groups

4 Retain or adopt on evidence

Old 1,713-row macro MAE primary
Cap64 FA/FR + P99 + worst underprediction
New 48 are secondary challenges

EVIDENCE CARDS / DIFFERENT COHORTS, NOT ONE ACCURACY TREND

V6 · fresh holdout

MAE −6.32%

192 newly measured circuits
Feature change within its grammar

V9 · consumed applications

MAE −25.40%

Examined application families
Development coverage improved

V12 · log target rejected

MAE +4.77%

Grouped 1,713-row OOF comparison
Raw target retained

V13 · targeted coverage

Old MAE +5.16%

New 48 MAE -94.65%
Not promoted · V9 retained

Retained V9: 1,713 labeled originals · 217 selected features · 300 trees. V13 improved its new coverage cohort but regressed the old primary; V9 stays retained.

Area of isolated original RTL only. These results do not establish CPU speed, routed timing, power, calibrated safety or independent application transfer. Full data/provenance: figure-data.json.